Chiara Segala

Curriculum Vitae

Personal Information

Date of Birth *April 30th, 1992*

Nationality Italian

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Employment

11/2024 – Present **Postdoc,** USI – Università della Svizzera Italiana, Dipartimento di Informatica e Facoltà di scienze economiche. Research group of Prof. Michael Multerer and Prof. Paul Schneider.

2022 – 2024 **Postdoc,** RWTH Aachen University, Institut für Geometrie und Praktische Mathematik. Research group of Prof. Michael Herty.

Education

2018 – 2022 **Ph.D. in Mathematics,** *Advanced numerical methods and control problems for multiscale models in self-organizing systems,* University of Trento.

Ph.D. Thesis: Robust control strategies for mean-field collective dynamics (**thesis defence on 30/06/2022**, with distinction).

Advisor: Prof. Giacomo Albi.

2017 - 2018 **Predoc Research Grant,** *Geometric control of mechanical system,* University of Verona.

Responsible: Prof. Riccardo Muradore and Prof. Nicola Sansonetto.

2014 – 2016 **Master in Mathematics, Modelling/Computational Curriculum,** University of Verona.

Thesis: Implementation of Exponential Integrators in GNU Octave.

Supervisor: Prof. Marco Caliari. Co-Supervisor: Prof. Alexander Ostermann (Innsbruck University).

2011 – 2013 **Bachelor in Mathematics, Economic/Financial Curriculum,** University of Verona.

Thesis: Forecasts and Statistical Analysis of Verona Import-Export.

Supervisor: Prof. Luca Di Persio.

Research Interests

Keywords **numerical analysis and optimization**

control (optimal control - model predictive control - sparse control - turnpike control)

multiagent systems (opinion dynamics - crowd dynamics - meanfield theory - multiscale models)

machine learning methods (neural networks - kernel methods - large scale approximation)

uncertainty quantification (Monte Carlo method - stochastic Galerkin - robust control)

mathematical finance (American option pricing - optimal stopping - conditional mean embeddings)

Affiliations (2019 – 2026) Gruppo Nazionale Calcolo Scientifico GNCS-INdAM.

Publications

Scopus link

submitted *Low-rank kernel methods for American option pricing*; M. Multerer, P. Schneider, C. Segala.

submitted *Tree-Adaptive Multiscale Kernel Lasso in Samplet Coordinates*; G. Fumagalli, M. Multerer, C. Segala.

accepted *Sparse stabilization of mean-field agent dynamics through a three-operator splitting gradient method*; G. Albi, D. Kalise, C. Segala, F. Zivcovich; **conference paper** in European Control Conferences (ECC26).

2026 *The turnpike control in stochastic multi-agent dynamics: a discrete-time approach with exponential integrators*; F. Cassini, C. Segala; **article** in Journal of Computational and Applied Mathematics (JCAM).

2025 *Sparse control in microscopic and mean-field leader-follower models*; M. Harms, M. Herty, C. Segala, E. Zerz; **article** in AIMS Mathematics.

2025 *Recent kernel methods for interacting particle systems: first numerical results*; C. Fiedler, M. Herty, C. Segala, S. Trimpe; **article** in European Journal of Applied Mathematics (EJAM).

2025 *Controllability of Continuous Networks and a Kernel-Based Learning Approximation*; M. Herty, C. Segala, G. Visconti; **book chapter** in Model Predictive Control: Engineering Methods for Economists.

2025 *Data/Moment-Driven Approaches for Fast Predictive Control of Collective Dynamics*; G. Albi, S. Bicego, M. Herty, Y. Huang, D. Kalise, C. Segala; **book chapter** in Model Predictive Control: Engineering Methods for Economists.

2024 *A Game-Theoretic Framework for Generic Second-Order Traffic Flow Models Using Mean Field Games and Adversarial Inverse Reinforcement Learning*; Z. Mo, X. Chen, X. Di, E. Iacomini, C. Segala , M. Herty, M. Lauriere; **article** in Transportation Science.

2024 *The turnpike property for high-dimensional interacting agent systems in discrete time*; M. Gugat, M. Herty, J. Liu, C. Segala; **article** in Optimal Control Applications and Methods.

2024 *The turnpike property for mean-field optimal control problems*; M. Gugat, M. Herty, C. Segala; **article** in European Journal of Applied Mathematics (EJAM).

2023 *Robust feedback stabilization of interacting multi-agent systems under uncertainty*; G. Albi, M. Herty, C. Segala; **article** in Applied Mathematics and Optimization (AMOP).

2023 *Reproducing kernel Hilbert spaces in the mean field limit*; C. Fiedler, M. Herty, M. Rom, C. Segala, S. Trimpe; **article** in Kinetic and Related Models (KRM).

2022 *Moment-Driven Predictive Control of Mean-Field Collective Dynamics*; G. Albi, M. Herty, D. Kalise, C. Segala; **article** in SIAM Journal on Control and Optimization (SICON).

2022 *Second-order optimal filter on TSE(2)*; R. Muradore, D. Rigo, N. Sansonetto, C. Segala; **article** in IEEE Transactions on Automatic Control.

2021 *Optimized leaders strategies for crowd evacuation in unknown environments with multiple exits*; G. Albi, F. Ferrarese, C. Segala; **book chapter** in Crowd Dynamics, Volume 3 (Modeling and Simulation in Science, Engineering and Technology).

2015 *Autoregressive approaches to import–export time series II: a concrete case study*; L. Di Persio, C. Segala; **article from bachelor thesis** in Modern Stochastics Theory and Applications.

Participation in Research Projects

2024 – Present Swiss National Science Foundation (SNSF 215528): *Large-scale kernel methods in financial economics*.

Responsible: Prof. Paul Schneider and Prof. Michael Multerer.

2022 – 2024 Project Sparsity and Singular Structures (SFB 1481): *Kinetic theory meets algebraic systems theory*.

Responsible: Prof. Michael Westdickenberg.

2021 – 2024 DFG Project Theoretical Foundations of Deep Learning (SPP 2298): *Assessment of Deep Learning through Meanfield Theory*.

Responsible: Prof. Gitta Kutyniok.

2019 Project GNCS-INDAM: *Approssimazione numerica di problemi di natura iperbolica e applicazioni*.

Responsible: Prof. Elisabetta Carlini.

2017 Project PRIN 2017: *Innovative Numerical Methods for Evolutionary Partial Differential Equations and Applications*.

Responsible: Prof. Giovanni Russo.

Research Funding and Support

2023 Received €3,000 in travel support via project SPP2298 for research visits and participation in scientific events.

- Additional funding received to support participation in scientific meetings (see **supported participation** below).

2012 Awarded a four-year scholarship by the University of Verona for bachelor and master studies.

Research Visits

2025 Research visiting, 28–31/07, Università degli Studi di Pavia, responsible Prof. Mattia Zanella.

2022 Research visiting with seminar talk, 07–12/11, Sapienza University of Rome, responsible Dr. Giuseppe Visconti.

2020 Research visiting, 06–11/01, RWTH Aachen University, responsible Prof. Michael Herty.

2019 Research visiting with seminar talk, 22–27/07, RWTH Aachen University, responsible Prof. Michael Herty.

2016 - 2017 Erasmus Traineeship at the University of Innsbruck (Austria), 12/2016–03/2017. Supervisor: Prof. Alexander Ostermann.

Other Experiences

2025-2026 External expert on the final examination board (Esame di Maturità), Liceo Lugano 3, Canton Ticino.

2016 Google Summer of Code, 01/05–31/08. Project under the supervision of Prof. Marco Caliari.

2014 Internship at the Chamber of Commerce of Verona, 01/04–30/06. Supervisor: Prof. Luca Di Persio.

Organizer of Scientific Meetings

2025 Mini-Symposium Organizer, Young Investigators Conference 2025, Pescara, Italy, 17–19/09.

2024 MFO Mini-Workshop High-Dimensional Control Problems and Mean-Field Equations with Applications in Machine Learning, Oberwolfach, 09–13/12.

2022 One day - Young Researcher Seminars Maths Applications and Models, 08/07, University of Verona.

2021 – 2022 Online Young Researcher Seminars Maths Applications and Models, University of Verona.

2019 INdAM GNCS Workshop on Numerical methods for multiscale control problems and applications, 07–08/02, University of Verona.

Communications and Participation in Scientific Events (selected)

2025 - NumPDE Summer Retreat 2025, Disentis/Mustér, Switzerland, 09–11/07.
(**Talk** *Moment-driven predictive control of mean-field collective dynamics*)

- Swiss Numerical Analysis Day 2025, University of Basel, 03/06.
(**Talk** *Moment-driven predictive control of mean-field collective dynamics*)

- DataHiking Workshop, RWTH Aachen University, 31/03–04/04.
(**Invited talk** *Control of continuous networks and a kernel-based learning approximation*, **supported participation**)

2024 - Warwick in Venice Workshop: New Trends in Optimal Control, Venezia, 15–17/05.
(**Invited talk** *Kernel methods for interacting particle systems: control surrogate modelling and mean-field limit*, **supported participation**)

- Workshop: Modeling, analysis, and control of multi-agent systems across scales, Pisa, 22–26/01.
(**Invited talk** *Kernel-based learning method for interacting particle systems and its mean field limit*, **supported participation**)

2023 - MFO Workshop: Control Methods in Hyperbolic Partial Differential Equations, Oberwolfach, 05–10/11.
(**Invited talk** *The turnpike property for mean-field optimal control problems*, **supported participation**)

- EUCCO 6th European Conference on Computational Optimization, Heidelberg, 25–27/09.
  (**Invited talk** *The turnpike property for mean-field optimal control problems*)
- ICIAM 2023, Tokyo, 20–25/08.
  (**Invited talk** *Kernel method for interacting particle systems and its mean field limit*)

2022 - Annual meeting: Theoretical Foundations of Deep Learning (SPP2298), Tutzing, 20–22/11.
  (**Talk** *Kernel method for interacting particle systems and its mean field limit*, **supported participation**)
- Workshop: Frontiers in numerical analysis of kinetic equations, Isaac Newton Institute for Mathematical Sciences Cambridge, 23–27/05.
  (**Poster** *Control strategies for collective dynamics with uncertainties*)

2021 - Asymptotic Behaviors of systems of PDEs arising in physics and biology, Polytech Lille, 16–19/11.
  (**Talk** *Robust control strategies for mean-field collective dynamics*)
- Dynamics Days, Nice, 22–29/08.
  (**Invited talk** *Moment-driven predictive control of mean-field collective dynamics*)

2020 - Summer School on Mathematical Physics, Ravello, 31/08–06/09.
  (**Talk** *Moment-driven predictive control of mean-field collective dynamics*, **supported participation**)
- Workshop GNCS Indam Project, Sapienza Roma, 06–07/02.
  (**Talk** *Mean-field feedback stabilization of collective behaviour*, **supported participation**)

2019 - Workshop: Feedback Control, RICAM Linz, 28–30/11.
  (**Poster** *Mean-field feedback stabilization of collective behaviour*, **supported participation**)
- Summer School: Trails in kinetic theory, Hausdorff Research Institute for Mathematics Bonn, 20–24/05.
  (**Poster** *Mean-field feedback stabilization of collective behaviour*, **supported participation**)

2018 - International Young Researchers Workshop on Geometry Mechanics and Control, University of Coimbra, 06–08/12.
  (**Poster** *Second-order optimal filter on TSE(2)*)
- Autumn School: From interacting Particle systems to Kinetic equations, University of Verona, 26–30/11.
  (**Talk** *Second-order optimal filter on TSE(2)*)

2017 - Octave Conference, CERN Geneva, 20–22/03.
  (**Talk** *Implementation of Exponential Integrators in GNU Octave*)

Computer Skills

Standard Python, Mathematica, `git`.
Advanced Matlab, Octave, LaTeX.

Languages

Italian Native speaker
English Fluent

Teaching Activities

2025 Assistant for "Mathematics for Data Science". USI – Università della Svizzera italiana, responsible Prof. Paul Schneider.

2024 Assistant for "Optimierung C". Mathematics master degree course, RWTH Aachen University, responsible Prof. Michael Herty.

2023 Assistant for "Optimierung A". Mathematics bachelor degree course, RWTH Aachen University, responsible Prof. Michael Herty.

2022 Assistant for "Optimierung C". Mathematics master degree course, RWTH Aachen University, responsible Prof. Michael Herty.

2022 Assistant for "Optimierung A". Mathematics bachelor degree course, RWTH Aachen University, responsible Prof. Michael Herty.

2020 Assistant for "Numerical Modelling and Optimization". Mathematics bachelor degree course, University of Verona, responsible Prof. Giacomo Albi.

2019 Assistant for "Functional Analysis". Mathematics master degree course, University of Verona, responsible Prof. Antonio Marigonda.

2018 Assistant for "Optimization". Mathematics master degree course, University of Verona, responsible Prof. Antonio Marigonda.

2018 Assistant for "Analisi Numerica II con Laboratorio". Mathematics bachelor degree course, University of Verona, responsible Prof. Marco Caliari.

Students Supervision

2023 Master thesis co-advisor of Ms. Jiehong Liu, RWTH Aachen University.

Reviewing Contributions

SIAM Journal on Control and Optimization, Journal of Hyperbolic Differential Equations (JHDE), Springer Journal of Computational and Applied Mathematics, Springer Journal of Optimization Theory and Applications (JOTA), IEEE Transactions on Signal and Information Processing over Network, 62nd IEEE Conference on Decision and Control (CDC 2023), Springer Journal of the Operations Research Society of China (JORC).

24.06.2026,